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Detection of Caries and Hypodontia using Deep Learning and Explainable AI

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Abstract—In today's world of technology breakthroughs, incorporating cutting-edge procedures has become essential, especially in healthcare, where an early and precise diagnosis is critical for improving patient outcomes and saving lives. Dental caries is an infectious bacterial disease primarily caused by the interaction of oral bacteria with dietary sugars, leading to plaque formation on teeth. Millions of people worldwide are facing this disease. Hypodontia is another dental term that healthcare providers use to describe congenitally missing teeth. We employed an AI-driven healthcare solution by utilizing a transformer-based image classification model, Vision Transformer (ViT), for the detection of dental caries and hypodontia. A dataset containing both periapical and panoramic dental x-rays utilized to gain maximum insights into teeth. The dataset was divided into 60% for training, 20% for testing, and 20% for validation. Data augmentation techniques are applied to enhance the diversity of training data and mitigate overfitting. The proposed model demonstrated remarkable performance and achieved an accuracy of 96.63% with evaluation matrices Precision, Recall, and F1-score of 96.60, 96.63, and 96.57 % respectively. Unlike Convolutional Neural Networks (CNN) which are black boxes, we utilized the Explainable Artificial Intelligence (XAI) models such as LIME, SHAP, and GRAD-CAM to gain insights into the inner workings of the ViT model. The XAI models contributed to fostering trust among medical dentists and medical professionals.

Index Terms—Dental Diseases, Caries Detection, Hypodontia, Vision Transformer, Explainable Artificial Intelligence (XAI)

I. INTRODUCTION

Dental health plays a vital role in life, with various conditions affecting millions worldwide [1]. By 2022, approximately 3.5 billion people were affected worldwide by oral diseases [2] [3]. Dental caries is the most prevalent disease globally known as tooth decay [4], is a multifaceted illness that begins with microbial alterations within the complex biofilm and is influenced by salivary flow and composition, fluoride exposure, sugar consumption, and preventive practices defined as tooth cleaning [5]. Another common dental condition, hypodontia refers to the absence of up to six teeth. Hypodontia may present in isolation or within genetic syndromes and

generally it carries a variable presentation of phenotypes. Both caries and hypodontia emphasize the need for early diagnosis before preventive dental care can be applied as shown in Fig. 1a and Fig. 1b. By automating detection, we can enhance accessibility to timely, preventive care in dental health.

Yet, there exists a limitation in terms of the requirement of human evaluation and the lack of models that can consider a set of conditions together. There are many approaches developed with machine learning for the detection of dental diseases. CNN models are mostly used for image classification but are often described as black-box models [6], due to their inability to explain the inner workings.

For instance, Lee et al. [7] obtained 65.02% of recall using a bitewing radiograph while Bayrakdar et al. [8] used the U-Net as well as VGG-16 models which achieved 81% of precision. These methods show promising results, but the subjectivity of manual assessment limits their reliability. A massive gap still exists for a completely automated model to appropriately diagnose both dental caries and hypodontia from an extensive range of imaging modalities. Hence, there is a need for innovative, comprehensive models that could enhance diagnostic accuracy and aid early intervention, thereby oral health care is more efficient [9]. Medical Images like panoramic and periapical are used as input for the model [10]. A diverse dataset can enhance feature extraction, contributing to higher accuracy in the detection of dental diseases. [11].

The main contributions of this paper are:

- Our proposed method improves the efficiency of the detection of caries and hypodontia.
- We utilized a diverse dataset containing both periapical and panoramic images to extract prominent features that enhance the model's decision-making process.

II. RELATED WORK

This section discusses the recent research on the multi-classification of dental diseases such as dental caries, hypodon-

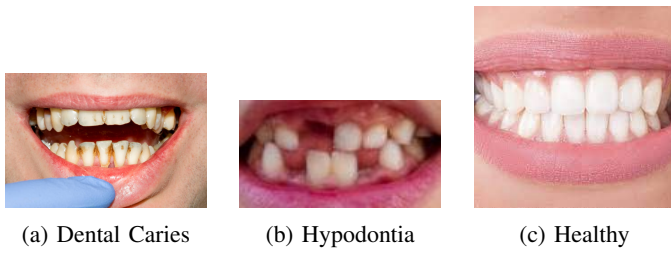


Fig. 1: Sample images from the dataset illustrating different classes

tia, and healthy teeth.

Park et al. [12] developed a deep learning model using different image classification and segmentation models such as U-Net, ResNet-18, and Faster-R-CNN, achieving an overall accuracy of 83.7%. The main limitation of this study is that intraoral photographic images cannot reveal internal or interproximal carious lesions as possible by X-ray images or tactile examinations, hence limiting the model's capacity to identify all caries types.

Kang et al. [13] utilized image-based deep learning techniques promising in the screening of dental caries using intraoral images. A total of 2,682 images and ensemble models such as Inception-ResNet-v2 were used which signifies its efficiency. Only advanced caries were recorded using ICDAS codes 4-6; no early lesions were recorded as ICDAS 1-3 have excluded them and limited the detection capabilities.

Esmailyfard et al. [11], developed a deep learning model for the classification of dental caries using Cone Beam Computed Tomography (CBCT) images. Data augmentation is applied to mitigate overfitting and increase the diversity of the training set. Multiple-input deep convolutional neural network architecture achieved an accuracy of 95.3% but the study lacks histologic gold standard, such as microcomputer tomography, for validating caries diagnosis.

Kawazu et al. [14] proposed that deep learning algorithms are helpful in dental diagnostics, particularly in detecting caries using intraoral and panoramic radiographs. The transfer learning approach which used pre-trained CNNs, have shown potential improvement with an accuracy of 87.9% having a small dataset size. The limitation includes the use of artificial caries images instead of real ones.

Inani et al. [15] employed an advanced deep learning model for binary classification of dental caries using various CNN models such as EfficientNetV2B3, VGG19, InceptionResNetV2, Xception, and ResNet50. Transfer learning techniques utilized, aiming to elevate the efficacy and precision of dental caries classification. Their proposed research concluded with 96.58% outperforming testing accuracy through VGG-19 over the existing models. More XAI models can also improve transparency.

Ayan et al. [16] found that the integration of Artificial Intelligence exhibits prominent and precise results in dental education and disease classification. Utilized a CNN model named "You Only Look Once" (YOLO) exhibited a marked

increase in their ability to find enamel and dental caries lesions in bitwise radiographs and achieved an accuracy of 89%. The use of images from a single institution database could introduce biases.

Research conducted by [17] that early diagnosis of dental caries progression can prevent invasive treatment. Deep learning techniques have gained prominence in the automatic analysis of dental radiographs. ResNet 50, ResNet 101, and MobileNetv3 models were used to attain a maximum testing accuracy of 96.37%, after data preprocessing. While internal tooth lesion detection is difficult using intraoral images.

TABLE I: Comparison of the proposed Vision Transformer (ViT) method for dental disease detection with existing state-of-the-art methods

Paper	Model	Dataset	Classification	F1	Acc. %
Park et al. 2022 [12]	U-Net, ResNet-18, Faster-R-CNN	2348 Intraoral Images	Affected / Normal	–	83.7
Qayyum et al. 2023 [17]	MobileNet, ResNet	Dental-caries	Caries / Normal	–	96.37
Kang et al. 2024 [13]	ResNet-50, Inception-v3, Inception-ResNet-v2, R-CNN	2682 Intraoral Images	Caries / Normal	–	87.1
Esmailyfard et al. 2024 [11]	CNN	785 Moral images	Caries / Normal	93.2	95.3
Kawazu et al. 2024 [14]	CNN	1094 Intraoral Images	Caries / Normal	–	87.9
Ayan et al. 2024 [16]	YOLO	1200 Bitewing Images	Caries / Normal	–	89.0
Inani et al. 2024 [15]	EffNetV2B3, VGG19, IncResNetV2, Xception, ResNet50	Teeth dataset	Caries / Normal	–	96.58
Proposed Work	ViT	Teeth dataset	Caries, Hypodontia, Normal	96.57	96.63

The comparative analysis is done in table I, which describes some of the recent developments in dental caries detection models, comparing them in terms of different architectures with different datasets and implying significant metrics such as classification accuracy and F1 scores.

III. MATERIALS AND METHODS

This section outlines the materials and methodologies used in the study for the multiclassification of dental diseases such as dental caries and hypodontia. The proposed methodology is composed of different subsections as follows.

A. Data Preprocessing

For the study, we utilized the teeth dataset, an open-source dataset available on Kaggle [18], along with additional images scraped from publicly available online resources. The dataset is divided into 5 classes: mouth ulcer, caries, hypodontia, tooth discoloration, and calculus. We used 2 classes from this such as caries and hypodontia. The dataset comprises high-resolution periapical and panoramic colored 1593 images, best suited for dental disease classification. The dataset was divided into 60% train, 20% test, and 20% validation. Data augmentation was utilized to artificially increase the diversity of the training dataset by changing the geometric and color configuration of the image. In healthcare, a minor error can result in severe consequences so analyzing the image from every angle is essential. We resized and randomly cropped the images to 224X224 pixels. The horizontal flip and rotation of images with 10 degrees were utilized in data augmentation. Regarding the picture resolution, the color jitter sets all the

values such as brightness, contrast, saturation, and hue to 0.1. Five classes are represented in the dataset but we utilized the critical and most important classes in this study. The dataset was mounted in the Google Colab environment by using function `mount.drive()` after being saved on Google Drive for quick access. To meet the Vision Transformer (ViT) input's image requirement, each image was scaled to 224 X 224 pixels as shown in the model name "vit-base-patch16-224-in21k".

B. Methodology

In recent years, deep learning has made important contributions to the disciplines of artificial intelligence and computer vision. CNNs have been the primary technique for image classification and object recognition tasks, excelling at localized feature extraction but frequently constrained by their localized receptive fields. However, medical imaging activities necessitate a more complete perspective to collect fine-grained information and global context for effective diagnosis. To meet these requirements, we used a cutting-edge Vision Transformer (ViT) model, a transformer-based technique that has shown improved performance in picture identification and classification tasks. Different data preprocessing steps were performed such as data augmentation, before passing to the model. Specifically, we utilized the "vit-base-patch16-224-in21k" model, a ViT version pre-trained on ImageNet-21k that processes images as fixed-size patches (16x16) at 224x224 pixels. This model is suitable for medium-sized datasets, however, larger ViT models are better for datasets with more than 50k images.

Unlike CNNs, which rely on convolutional filters for localized feature extraction, the ViT model uses a self-attention mechanism to capture relationships throughout the image. Our method separated images into patches, which were then fed via a transformer encoder after being positionally encoded. Layer normalization, multi-head attention, and multi-layer perception procedures all contributed to excellent feature extraction in this encoder. The final classification was carried out by a feed-forward network that used a softmax function to generate class probabilities. To promote openness in decision-making, we used the XAI approach to make our model's predictions interpretable.

Algorithm 1 Multi-Classification of Dental Diseases Using Vision Transformer

```

1: Input: Image dataset of DXR =  $\{I_1, I_2, \dots, I_n\}$ 
2: Load pre-trained ViTForImageClassificationModel
3: Initialize model components
4:  $vit \leftarrow ViTModel$ 
5:  $embeddings \leftarrow ViTEmbeddings$ 
6:  $patch\_embeddings \leftarrow ViTPatchEmbeddings$ 
7:  $projection \leftarrow Conv2d(3, 768, kernel\_size = (16, 16), stride = (16, 16))$ 
8:  $dropout \leftarrow Dropout(p = 0.0, inplace = False)$ 
9:  $encoder \leftarrow ViTEncoder$ 
10: for layer  $\leftarrow 1$  to 12 do
11:    $attention \leftarrow ViTSdpaAttention$ 
12:    $attention\_block \leftarrow ViTSdpaSelfAttention$ 
13:    $query, key, value \leftarrow Linear(in\_features = 768, out\_features = 768, bias = True)$ 
14:    $dropout \leftarrow Dropout(p = 0.0, inplace = False)$ 
15:    $output\_block \leftarrow ViTSelfOutput$ 
16:    $dense \leftarrow Linear(in\_features = 768, out\_features = 768, bias = True)$ 
17:    $dropout \leftarrow Dropout(p = 0.0, inplace = False)$ 
18: end for
19:  $intermediate\_block \leftarrow ViTIntermediate$ 
20:  $dense \leftarrow Linear(in\_features = 768, out\_features = 3072, bias = True)$ 
21:  $activation \leftarrow GELUActivation()$ 
22:  $output\_block \leftarrow ViTOutput$ 
23:  $dense \leftarrow Linear(in\_features = 3072, out\_features = 768, bias = True)$ 
24:  $dropout \leftarrow Dropout(p = 0.0, inplace = False)$ 
25: Add Classifier
26:  $classifier \leftarrow Linear(in\_features = 768, out\_features = 5, bias = True)$ 
27: Classification scores for each input image = 0

```

XAI is essential, especially in healthcare, where understanding the basis for a model's predictions can enhance trust, reliability, and acceptance among practitioners. Our study used model-agnostic XAI techniques, including LIME [19], SHAP [20], and Grad-CAM [21], to visualize and interpret the influential factors behind each prediction. These techniques offer insight into the regions of the input image that were most critical to the model's decision, helping to elucidate the "black box" [6] nature of deep learning models. This methodology is illustrated in Fig. 2, which shows the ViT architecture with patch embeddings, attention mechanisms, and transformer blocks, as well as the integration of XAI techniques to aid in interpretation. For reproducibility and future model improvements, we saved the trained model as a pickle file. The above algorithm explains the vision transformer architecture such as patch embeddings, attention mechanism, convolutional layers, drop-out layers, and layer normalization.

IV. RESULTS AND DISCUSSION

The experiments and the model training in this research were conducted on the Google Colab environment, which provides access to cloud-based GPUs for high-performance computation. The environmental specifications include a GPU, NVIDIA Tesla T4, a Central Processing Unit (CPU) Intel Core

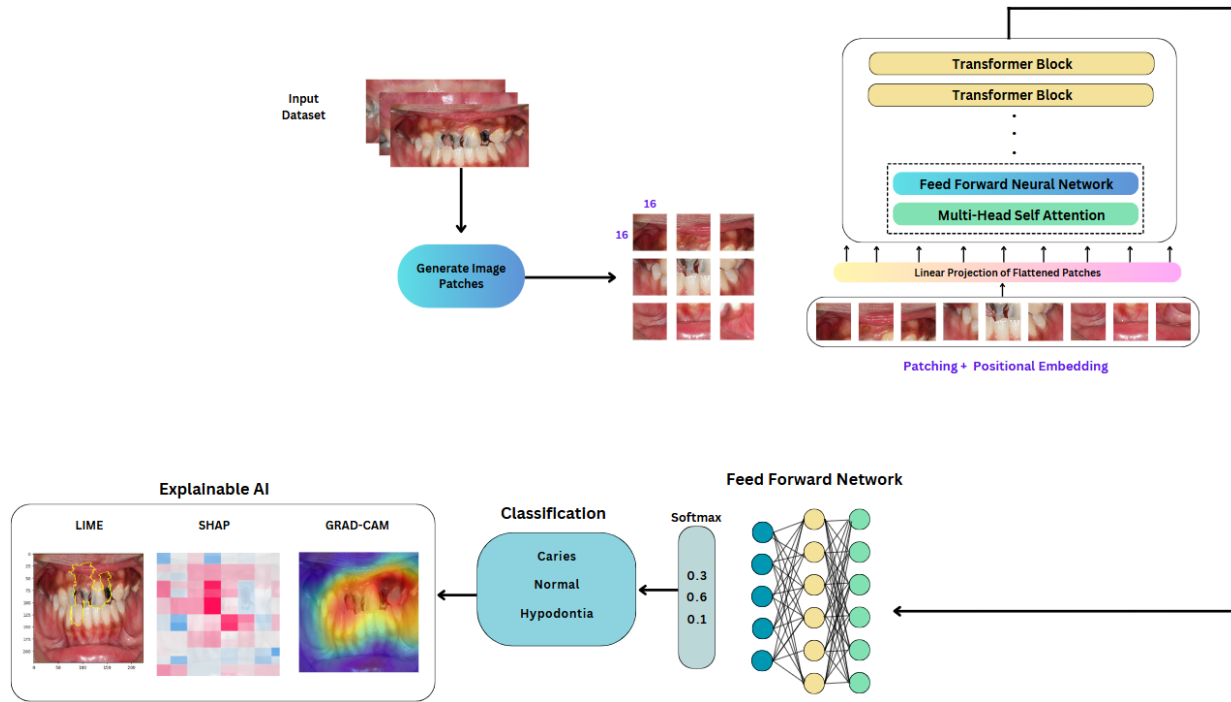


Fig. 2: Proposed system architecture consists of image patch generation, feature extraction via vision transformer, classification using feed-forward network, and explainable AI.

i5-6200 with 8GB of RAM, and a 256GB SSD, operating on Windows 10 with Python Version 3.12.4.

Various parameters were discussed to gain insights into the results of the proposed model, including evaluation metrics such as Accuracy, Recall, F1-Score, and Precision.

- **Accuracy:** The accuracy of the model is calculated as the ratio of correctly predicted instances to the total instances.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision:** Precision refers to the accuracy of the model's positive predictions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

- **F1-Score:** The F1-Score is used to measure the overall performance of the model, as it is the harmonic mean of Precision and Recall.

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- **Recall:** Recall measures the effectiveness of the model in identifying positive instances in the complete dataset.

$$\text{Recall} = \frac{TP}{TP + FN}$$

The confusion matrix summarizes the performance of the proposed model based on test data, showing accurate and inaccurate instances of model prediction. The matrix is shown in Fig. 3 with true and predicted labels.

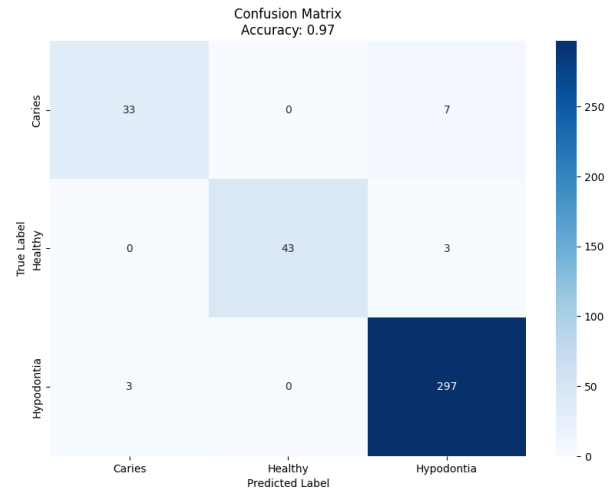


Fig. 3: Confusion Matrix showing the performance of the proposed model for the classification of Caries, Healthy, and Hypodontia categories with detailed true positive, false positive, and false negative distributions.

The given confusion matrix reflects the performance of the classification model applied to three classes: Caries, Healthy, and Hypodontia. All classes are correctly classified at the accuracy level of 96.63%. For the caries class, out of 40 actual cases, it happened that the model correctly classified 33 with 7 classified as hypodontia. In the Healthy class, it classified 43 of the 46 cases and misclassified 3 instances as

hypodontia. For Hypodontia, it excellently identified all but 3 cases correctly out of 300, misclassifying one as caries. At the initial phase of model training, the model learns well from the training data, the training loss decrease corresponds to validation loss. The model is trained on 7 epochs and achieved an accuracy of 96.63% as seen in Fig. 4 with hyperparameters as the batch size was 32 and learning rate 0.01. Further increasing the number of epochs to 80 did not improve the accuracy. For hyperparameter tuning, a batch size of 16 was used with different learning rates as 0.001 and 0.1 but the model always achieved an accuracy of less than 96%. During

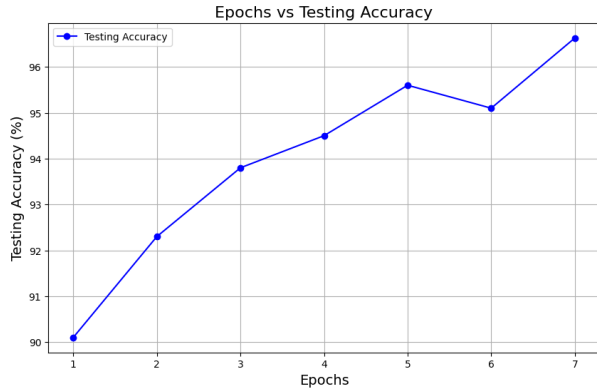


Fig. 4: Testing accuracy trend of the proposed model over the epochs, illustrating consistent enhancements in performance.

the requirement analysis of the proposed model, the vision transformer was chosen based on several factors. The dataset used contains nearly 1600 images, so the base ViT is better to accommodate the whole process. However different ViT models can be utilized based on data like ViT-B/32, the base model with an input patch size of 32 X 32, ViT-L/16, ViT-L/32, and ViT-H/14 are large variants of base ViT models. They require a large amount of data for efficient and robust performance. Initially, we implemented a large variant of ViT due to robust performance and excellent accuracy.

XAI models like LIME and SHAP show excellent performance by highlighting the region that contributes to the positive and negative prediction of the model implied via the ViT. Still, the GAD-CAM could not be completely implemented. However, we used another pre-trained deep learning model named ResNet50, and the GRAD-CAM was implemented successfully, which generated the heatmap to highlight the affected region.

- **LIME (Local Interpretable Model-Agnostic Explanations):** LIME emphasizes the importance of explanations in machine learning models to build trust and transparency [19]. It uses a perturbation-based technique to explain complex black-box models by providing explanations.

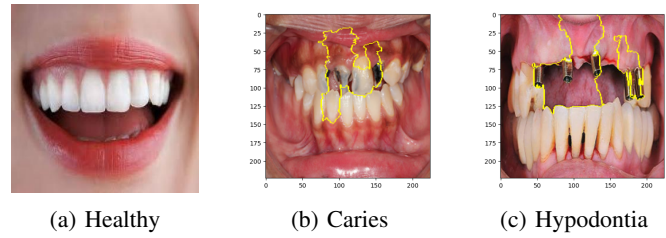


Fig. 5: LIME visualizations highlighting feature importance for classifying dental conditions. The images show the marked boundaries.

As shown in Fig. 5, different dental disease images were passed through the LIME model after classification from the ViT model. The LIME model marks the boundaries of the dental X-ray images for Caries and Hypodontia.

- **SHAP (Shapley Additive Explanations):** SHAP is based on cooperative game theory and provides consistent and accurate feature importance scores [20]. SHAP generates scatterplots where dots represent a feature's SHAP value for an instance. Red dots indicate a positive contribution, and blue dots indicate a negative contribution as visualized in Fig. 6b and Fig. 6c.

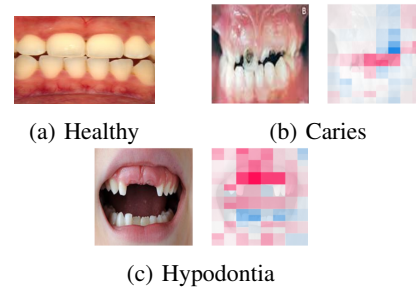


Fig. 6: SHAP visualizations showcasing feature contributions for multi-classification of dental conditions.

- **GRAD-CAM (Gradient Weighted Class Activation Mapping):** Grad-CAM is a technique used to generate heatmaps that provide visual explanations for CNN-based model decisions. The Grad-CAM heatmap was used to analyze images of different dental diseases after classification by the ViT model, with the affected region highlighted in red as demonstrated in Fig. 7b and Fig. 7c.

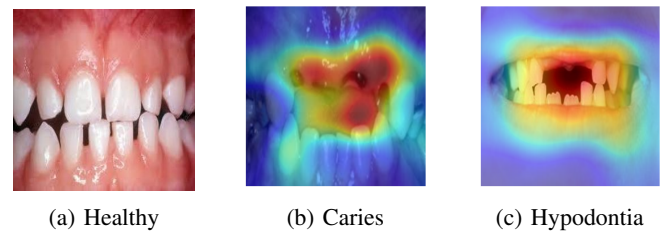


Fig. 7: Grad-CAM visualizations with heatmaps for the classification of dental diseases.

TABLE II: Comparison of results of the proposed study with existing state-of-the-art methods for dental disease classification.

Paper	Class.	Meth.	Aec.	Prec.	Rec.	F1	XAI
Lee et al. [7]	Caries/Normal	U-Net	-	63.3	65.0	64.1	No
Bayrakdar et al. [22]	Caries/Normal	VGG-16, U-Net	-	84.0	84.0	84.0	No
Thanh et al. [23]	Caries/Normal	Faster R-CNN, YOLOv3	87.4	77.3	71.2	-	No
Zhu et al. [24]	Caries/Normal	CNN	93.6	94.1	86.0	92.6	No
Tareq et al. [25]	Caries/Normal	YOLO, VGG16	87.0	89.0	88.0	88.0	No
Ciftci et al. [26]	Caries/Normal	DT, RF, MLP	95.8	96.0	96.0	96.0	No
Proposed Model	Caries, Hypodontia, Normal	ViT	96.63	96.60	96.63	96.57	Yes

A method proposed for detecting dental caries and hypodontia, based on a ViT and XAI, also performed better in this challenge than the existing models in the literature. Table II exhibits the superior performance of the proposed model over the existing study. In addition, the role of XAI in assuring model transparency is vital. This is because the acceptance of the models by the users is the most important prerequisite. Such acceptance is crucial for the deployment of any black box AI model or application in medical diagnosis.

V. CONCLUSION

Dental caries and hypodontia are common conditions that are misdiagnosed, highlighting the need for better diagnostic methods. In this study, we created an AI-driven healthcare solution for multi-classification based on dental X-rays, with data augmentation to improve model variability. The suggested model achieved a strong accuracy of 96.63%, backed by good precision, recall, and F1 score. To promote confidence and transparency in decision-making, we used XAI tools such as LIME, SHAP, and GRAD-CAM, which provide interpretable insights for practitioners. The collaboration between ViT and XAI is a key step toward advancing AI diagnostics in dentistry. Future research will look at applying this method to other dental diseases.

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